

Generalized Harmonic Progression Part II

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Abstract

In a previous paper, we saw how to create formulae for the sum of the terms of a harmonic progression of order k , $HP_k(n)$, with integer parameters, a and b . In this new paper we make those formulae even more general by lifting the restriction that the parameters be integers. The new formula holds always, except when $ib/a \in \mathbb{Z}$. Here a slightly modified version of the reasoning used before is presented, in a standalone exposition that does not require prior knowledge of the precursor paper. The limit of $HP_k(n)$ when n approaches infinity is also found, giving a new formula for the Hurwitz zeta function.

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1 Introduction

This article demonstrates how to produce a formula for a harmonic progression of order k with complex parameters,

$$HP_k(n) = \sum_{j=1}^n \frac{1}{(a\mathbf{i}j + b)^k},$$

with a an integer, b a complex number and $\mathbf{i}$ the imaginary unit.

The motivation to create such a formula came from the realization that, unlike the sum of $1/(j^2 - 1)$ over j , a sum such as $1/(j^2 + 1)$ can not be derived from the formulae created in [4].

If a formula for $HP_k(n)$ with integer a and complex b can be produced, formulae for most, if not all, sums of the type $\sum_j 1/p(j)$, where $p(j)$ is a polynomial with complex coefficients, can be produced. Besides, $HP_k(n)$ provides the partial sums of the Hurwitz zeta function, $\zeta(k, b)$, at the positive integers k , whose formula can be obtained by taking the limit of $HP_k(n)$ when n goes to infinity. With a little more effort, it might even be possible to obtain the analytic continuation of $\zeta(k, b)$.

The choice of integer a and complex b is good enough in most of the cases (since one can always make a integer by dividing the parameters by a), and when it is not, one can resort to the formulae from the previous paper [4], which assume a and b to be integers.

The method makes use again of the Faulhaber formula [2] for the sum of the powers of the first n positive integers, which are given by,

$$\sum_{j=1}^n j^{2i} = \frac{n^{2i+1}}{2i+1} + \sum_{j=0}^i \frac{(2i+1)! B_{2j}}{(2j)!(2i+1-2j)!} n^{2i+1-2j} \quad (1)$$

$$\sum_{j=1}^n j^{2i+1} = \frac{n^{2i+2}}{2i+2} + \sum_{j=0}^i \frac{(2i+2)! B_{2j}}{(2j)!(2i+2-2j)!} n^{2i+2-2j}, \quad (2)$$

where B_{2j} are the non-zero Bernoulli numbers.

2 Review of preview results

Before we begin, let us recap some results from [4] for a bit of context. We had seen that the sum of the terms of a generalized harmonic progression with integer parameters,

a and b , for even and odd powers is given, respectively, by,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(aj+b)^{2k}} &= -\frac{1}{2b^{2k}} + \frac{1}{2(an+b)^{2k}} \\ &- \frac{(-1)^k (2\pi)^{2k}}{2} \int_0^1 \sum_{j=0}^k \frac{B_{2j} (2-2^{2j}) (1-u)^{2k-2j}}{(2j)!(2k-2j)!} (\sin 2\pi(an+b)u - \sin 2\pi bu) \cot \pi au \, du, \\ \sum_{j=1}^n \frac{1}{(aj+b)^{2k+1}} &= -\frac{1}{2b^{2k+1}} + \frac{1}{2(an+b)^{2k+1}} \\ &- \frac{(-1)^k (2\pi)^{2k+1}}{2} \int_0^1 \sum_{j=0}^k \frac{B_{2j} (2-2^{2j}) (1-u)^{2k+1-2j}}{(2j)!(2k+1-2j)!} (\cos 2\pi(an+b)u - \cos 2\pi bu) \cot \pi au \, du \end{aligned}$$

Notice that there can be singularities on both sides of these equations if, for example, $aj+b=0$ for some $j \leq n$ (that is, a divides b and they have different signs). In those cases, if the singularities are ignored wherever they occur the formula still holds.

This means that, for instance, if $b=0$ then the above formulae reduce to the the generalized harmonic numbers formulae from [3].

To see why singularities can be removed, let us look at the formula for the odd powers. From that formula, it is easy to see why it still holds when singularities are ignored. For example, let us take the forward difference of $HP(n)$,

$$2\pi \int_0^1 (1-u) (\cos 2\pi(an+b)u - \cos 2\pi(a(n-1)+b)u) \cot \pi a(1-u) \, du = -\frac{1}{an+b} - \frac{1}{a(n-1)+b}$$

Now, if for instance $an+b=0$, the left-hand side of the above equation yields $1/a$, even though $1/(an+b)$ is a singularity on the right-hand side. Therefore, by ignoring the singularity, the equation still holds.

There was also a unified formula based on the exponential. For every integer a , b and $k \geq 1$,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(aj+b)^k} &= -\frac{1}{2b^k} + \frac{1}{2(an+b)^k} \\ &+ \frac{i(2\pi i)^k}{2} \int_0^1 \sum_{j=0}^k \frac{B_j (1-u)^{k-j}}{j!(k-j)!} (e^{2\pi i(an+b)u} - e^{2\pi i b u}) \cot \pi au \, du \quad (3) \end{aligned}$$

3 Approach based on exponential

The reasoning to build a formula for the harmonic progression relies on the Taylor series expansion of the exponential, and seizes upon the fact that it is constant for any integer a and complex b ,

$$e^{2\pi(aij+b)} = e^{2\pi b} \Rightarrow \sum_{i=0}^{\infty} \frac{(2\pi(aij+b))^i}{i!} = e^{2\pi b}, \text{ if } a \text{ is integer}$$

Starting from the above initial equation, one ends up with the below after a few steps,

$$\frac{1}{aij+b}(e^{2\pi b} - 1) = 2\pi \sum_{i=0}^{\infty} \left(\frac{(2\pi(aij+b))^{2i}}{(2i+1)!} + \frac{(2\pi(aij+b))^{2i+1}}{(2i+2)!} \right) \quad (4)$$

Note there are now two power series, as opposed to one previously.

At this point it is evident that this formula does not work if $e^{2\pi b} = 1$, which happens when b is a purely complex number with integer imaginary part ($\Re(b) = 0$ and $\Im(b) \in \mathbb{Z}$). In most situations, this issue can be avoided by making b non-integer, such as by dividing b by a . Likewise, the formula will not apply if a is not integer ($a \notin \mathbb{Z}$). In that case a can be made integer by dividing both parameters by a ,

$$\sum_{j=1}^n \frac{1}{aij+b} = \frac{1}{a} \sum_{j=1}^n \frac{1}{ij+b/a}$$

However, in both cases this work-around will not work if ib/a is integer, including zero. In those cases the previous formulae are applicable.

3.1 Lagrange's identities

Through Lagrange's identities, which stem from the sum of the terms of a geometric progression, functions can be created whose power series take similar forms to the power series that are obtained when the initial equation from the previous section is expanded.

For the complex harmonic progression, these functions are the same as the ones used in [4], with only a slight transformation ($a := ai$). Nonetheless, they are obtained by means of the Lagrange's trigonometric identities,

$$\begin{aligned} \sum_{j=1}^k \cos \frac{2\pi n(aij+b)}{k} &= -\frac{1}{2} \cos \frac{2\pi bn}{k} + \frac{1}{2} \cos 2\pi n \left(ai + \frac{b}{k} \right) + \cos \pi n \left(ai + \frac{2b}{k} \right) \sin \pi ain \cot \frac{\pi ain}{k} \\ \sum_{j=1}^k \sin \frac{2\pi n(aij+b)}{k} &= -\frac{1}{2} \sin \frac{2\pi bn}{k} + \frac{1}{2} \sin 2\pi n \left(ai + \frac{b}{k} \right) + \sin \pi n \left(ai + \frac{2b}{k} \right) \sin \pi ain \cot \frac{\pi ain}{k}, \end{aligned}$$

along with the identities,

$$\cos(x + y) = \cos x \cos y - \sin x \sin y, \text{ and } \sin(x + y) = \sin x \cos y + \cos x \sin y$$

Power series for the left-hand side of the Lagrange identities can be derived with the employment of (1) and (2), resulting in the following power series for each function on the right-hand side,

$$\cos 2\pi n \left(ai + \frac{b}{k} \right) = \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i} \left(\sum_{j=0}^i \frac{(aik/b)^{2j}}{(2j)!(2i-2j)!} + \sum_{j=0}^{i-1} \frac{(aik/b)^{2j+1}}{(2j+1)!(2i-2j-1)!} \right)$$

$$\sin 2\pi n \left(ai + \frac{b}{k} \right) = \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i+1} \left(\sum_{j=0}^i \frac{(aik/b)^{2j}}{(2j)!(2i+1-2j)!} + \sum_{j=0}^i \frac{(aik/b)^{2j+1}}{(2j+1)!(2i-2j)!} \right)$$

$$\begin{aligned} & \cos \pi n \left(ai + \frac{2b}{k} \right) \sin \pi ain \cot \frac{\pi ain}{k} = \\ & \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i} \left(\sum_{j=0}^i \frac{(ai/b)^{2j}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^{i-1} \frac{(ai/b)^{2j+1}}{(2i-2j-1)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \\ & \sin \pi n \left(ai + \frac{2b}{k} \right) \sin \pi ain \cot \frac{\pi ain}{k} = \\ & \sum_{i=0}^{\infty} (-1)^i \left(\frac{2\pi bn}{k} \right)^{2i+1} \left(\sum_{j=0}^i \frac{(ai/b)^{2j}}{(2i+1-2j)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^i \frac{(ai/b)^{2j+1}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} k^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \end{aligned}$$

3.2 Harmonic progression

Going back to equation (4), it is summed over j , and then $(aij + b)^{2i}$ and $(aij + b)^{2i+1}$ are expanded using the binomial theorem, and lastly the sums of j^{2q} and j^{2q+1} over j are replaced with their respective Faulhaber formulae. After all the possible simplifications are done, one ends up with a few power series.

First one has the independent term given by,

$$-\pi \sum_{i=0}^{\infty} \left(\frac{(2\pi b)^{2i}}{(2i+1)!} + \frac{(2\pi b)^{2i+1}}{(2i+2)!} \right) = -\frac{e^{2\pi b} - 1}{2b}$$

For the other, more complicated power series, integral representations can be obtained with the aid of their associated functions from (3.1),

$$\begin{aligned}
\pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i}}{2i+1} \left(\sum_{j=0}^i \frac{(ain/b)^{2j}}{(2j)!(2i-2j)!} + \sum_{j=0}^{i-1} \frac{(ain/b)^{2j+1}}{(2j+1)!(2i-2j-1)!} \right) &= -\frac{i\pi}{n} \int_0^{in} \cos 2\pi x \left(ai + \frac{b}{n} \right) dx \\
\pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i+1}}{2i+2} \left(\sum_{j=0}^i \frac{(ain/b)^{2j}}{(2j)!(2i+1-2j)!} + \sum_{j=0}^i \frac{(ain/b)^{2j+1}}{(2j+1)!(2i-2j)!} \right) &= -\frac{\pi}{n} \int_0^{in} \sin 2\pi x \left(ai + \frac{b}{n} \right) dx \\
2\pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i}}{2i+1} \left(\sum_{j=0}^i \frac{(ai/b)^{2j}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^{i-1} \frac{(ai/b)^{2j+1}}{(2i-2j-1)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \\
&= -\frac{2\pi i}{n} \int_0^{in} \cos \pi x \left(ai + \frac{2b}{n} \right) \sin(\pi a i x) \cot \frac{\pi a i x}{n} dx \\
2\pi \sum_{i=0}^{\infty} \frac{(2\pi b)^{2i+1}}{2i+2} \left(\sum_{j=0}^i \frac{(ai/b)^{2j}}{(2i+1-2j)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+1-2q}}{(2j+1-2q)!(2q)!} + \sum_{j=0}^i \frac{(ai/b)^{2j+1}}{(2i-2j)!} \sum_{q=0}^j \frac{B_{2q} n^{2j+2-2q}}{(2j+2-2q)!(2q)!} \right) \\
&= -\frac{2\pi}{n} \int_0^{in} \sin \pi x \left(ai + \frac{2b}{n} \right) \sin(\pi a i x) \cot \frac{\pi a i x}{n} dx
\end{aligned}$$

Hence, by putting it all together and making a change of variables in the integral, the below formula is found, which works as long as a is integer and ib is not integer,

$$\sum_{j=1}^n \frac{1}{aij+b} = -\frac{1}{2b} + \frac{1}{2(ain+b)} + \frac{2\pi}{e^{2\pi b} - 1} \int_0^1 e^{\pi(ain+2b)u} \sin \pi a n u \cot \pi a u du$$

In this formula, the independent term came from the first power series, the second term from the next two power series and the third term from the last two power series, which is somewhat evident.

As discussed before, provided that $ib/a \notin \mathbb{Z}$, the restriction that a be integer can be lifted by transforming the formula, for example, as follows,

$$\frac{1}{a} \sum_{j=1}^n \frac{1}{ij+b/a} = -\frac{1}{2b} + \frac{1}{2(ain+b)} + \frac{2\pi}{a(e^{2\pi b/a} - 1)} \int_0^1 e^{\pi(in+2b/a)u} \sin \pi n u \cot \pi u du,$$

which holds for any $a, b \in \mathbb{C}$. There are endless ways of doing that, the example is just one of the possibilities.

3.3 Generalization

To generalize the previous formula created, first note that a recursion is obtained at each new step. For example, for the third power one has,

$$(e^{2\pi b} - 1) \sum_{j=1}^n \frac{1}{(a\mathbf{i}j + b)^3} = \sum_{j=1}^n \left(\frac{(2\pi)^2}{2!} \frac{1}{a\mathbf{i}j + b} + \frac{2\pi}{1!} \frac{1}{(a\mathbf{i}j + b)^2} + (2\pi)^3 \sum_{i=0}^{\infty} \frac{(2\pi(a\mathbf{i}j + b))^{2i}}{(2i+3)!} + \frac{(2\pi(a\mathbf{i}j + b))^{2i+1}}{(2i+4)!} \right) \quad (5)$$

That means that only the last sum in the above equation is complicated. As before, this recurrence is such that in the final formula the terms that go outside of the integral reduce to very simple forms, with the integral being the only challenging part.

Let $p_k(u)$ be the polynomial in u that goes within the integral. Then for integer a and non-integer $\mathbf{i}b/a$,

$$\sum_{j=1}^n \frac{1}{(a\mathbf{i}j + b)^k} = -\frac{1}{2b^k} + \frac{1}{2(a\mathbf{i}n + b)^k} + (2\pi)^k \int_0^1 p_k(u) e^{\pi(a\mathbf{i}n+2b)u} \sin \pi a n u \cot \pi a u \, du, \quad (6)$$

where $p_k(u)$ is given by the recurrence equation,

$$(e^{2\pi b} - 1)p_k(u) = \begin{cases} 1, & \text{if } k = 1 \\ \frac{(1-u)^{k-1}}{(k-1)!} + \sum_{j=1}^{k-1} \frac{p_j(u)}{(k-j)!}, & \text{if integer } k > 1 \end{cases}$$

and where term $(1-u)^{k-1}/(k-1)!$ came from the observation of the patterns.

Now to solve the recurrence, let $p(x)$ be the generating function of $p_k(u)$. By analyzing the recurrence, it is possible to conclude that,

$$(e^{2\pi b} - 1)p(x) - (-1 + e^x)p(x) = x e^{(1-u)x} \Rightarrow p(x) = -\frac{x e^{(1-u)x}}{e^x - e^{2\pi b}}$$

The solution of this equation is not trivial, as the general term of the power series of $p(x)$ is not simple.

To find $p_k(u)$, let δ_{ij} be the Kronecker delta ($\delta_{ij} = 1$, if $i = j$, $\delta_{ij} = 0$ otherwise), and $\text{Li}_k(z)$ the polylogarithm, a function given by a Dirichlet series,

$$\text{Li}_k(z) = \sum_{j=1}^{\infty} \frac{z^j}{j^k},$$

which has an analytic continuation for $|z| > 1$.

This function provides the coefficients of the Taylor series expansion of $p(x)$, that is,

$$p_k(u) = \frac{p^{(k)}(0)}{k!} = \sum_{j=1}^k \frac{(\delta_{1j} + \text{Li}_{-j+1}(e^{-2\pi b})) (1-u)^{k-j}}{(j-1)!(k-j)!},$$

and the below identity,

$$\sum_{j=1}^k \frac{\text{Li}_{-j+1}(e^{-2\pi b})(1-u)^{k-j}}{(j-1)!(k-j)!} = \frac{e^{-2\pi b}}{\Gamma(k)} \Phi(e^{-2\pi b}, -k+1, 2-u),$$

provides an extension of the formula to non-integer k .

A simplification can be attained through this identity that stems from the definition of the Lerch Φ function,

$$(1-u)^{k-1} + e^{-2\pi b} \Phi(e^{-2\pi b}, -k+1, 2-u) = \Phi(e^{-2\pi b}, -k+1, 1-u), \quad (7)$$

and also Euler's formula,

$$e^{\pi i a n u} \sin \pi a n u = (\cos \pi a n u + i \sin \pi a n u) \sin \pi a n u = \frac{i}{2} (1 - e^{2\pi i a n u}), \quad (8)$$

Hence, if $\Re(k) > 0$ and a is integer, one has,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(a i j + b)^k} &= -\frac{1}{2b^k} + \frac{1}{2(a i n + b)^k} \\ &\quad - \frac{i(2\pi)^k}{2\Gamma(k)} \int_0^1 e^{-2\pi b(1-u)} \Phi(e^{-2\pi b}, -k+1, 1-u) (e^{2\pi i a n u} - 1) \cot \pi a u \, du \end{aligned} \quad (9)$$

Finally, the formula can be turned into a neater form by making the offset parameter equal to $i b$,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(a j + b)^k} &= -\frac{1}{2b^k} + \frac{1}{2(a n + b)^k} \\ &\quad - \frac{i(2\pi i)^k}{2\Gamma(k)} \int_0^1 e^{-2\pi i b(1-u)} \Phi(e^{-2\pi i b}, -k+1, 1-u) (e^{2\pi i a n u} - 1) \cot \pi a u \, du \end{aligned} \quad (10)$$

3.4 Solution of the transform

The below transform of (6), which is simply setting $a := 1$ and $b := b/a$ and multiplying both sides by a^{-k} , is more useful for the purpose of finding out the limit of the harmonic progression,

$$\sum_{j=1}^n \frac{1}{(aj+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(ain+b)^k} + \left(\frac{2\pi}{a}\right)^k \int_0^1 p_k(u) e^{\pi(in+2b/a)u} \sin \pi nu \cot \pi u \, du \quad (11)$$

Now the generating function of the $p_k(u)$ becomes,

$$p(x) = -\frac{x e^{x(1-u)/a}}{a(e^{x/a} - e^{2\pi b/a})} \Rightarrow \frac{p^{(k)}(0)}{k!} = \frac{e^{-2\pi b/a}}{a^k(k-1)!} \Phi(e^{-2\pi b/a}, -k+1, 1-u),$$

Therefore, if $\Re(k) > 0$ and $ib/a \notin \mathbb{Z}$,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(aj+b)^k} &= -\frac{1}{2b^k} + \frac{1}{2(ain+b)^k} \\ &\quad - \frac{i}{2\Gamma(k)} \left(\frac{2\pi}{a}\right)^k \int_0^1 e^{-2\pi(1-u)b/a} \Phi(e^{-2\pi b/a}, -k+1, 1-u) (e^{2\pi i nu} - 1) \cot \pi u \, du \end{aligned} \quad (12)$$

The formula can be simplified further by setting $a = -i$, and thus if $\Re(k) > 0$ and $b \notin \mathbb{Z}$,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^k} &= -\frac{1}{2b^k} + \frac{1}{2(n+b)^k} \\ &\quad - \frac{i(2\pi i)^k}{2\Gamma(k)} \int_0^1 e^{-2\pi i b(1-u)} \Phi(e^{-2\pi i b}, -k+1, 1-u) (e^{2\pi i nu} - 1) \cot \pi u \, du \end{aligned} \quad (13)$$

3.5 Application examples

Let us see two examples on how the newfound formulae can be applied to obtain sums that were not possible with the formulae created in the previous paper^[4].

The first example is the sum of $1/(j^2 + 1)$ over j . The first step to derive this formula is to decompose it into a combination like,

$$\frac{1}{j^2 + 1} = \frac{1}{2(ij + 1)} - \frac{1}{2(ij - 1)}$$

The below formula is found after a few simplifications,

$$\sum_{j=1}^n \frac{1}{j^2 + 1} = -\frac{1}{2} + \frac{1}{2(n^2 + 1)} + \pi \operatorname{csch} 2\pi \int_0^1 \cosh 2\pi(2u - 1) \sin 4\pi n u \cot 2\pi u \, du$$

Generally, to obtain the formula for the sum of $1/(j^{2k} + 1)$ over j it is necessary to find all the $2k$ complex roots of $x^{2k} = -1$ (say they are x_j), and coefficients c_j such that,

$$\sum_{j=1}^{2k} \frac{c_j}{x - x_j} = \frac{1}{x^{2k} + 1}$$

The second example is another quadratic polynomial, which can be decomposed as below,

$$\frac{1}{j^2 + 2j + 2} = \frac{1}{2(\mathbf{i}j + 1 + \mathbf{i})} - \frac{1}{2(\mathbf{i}j - 1 + \mathbf{i})},$$

and which therefore implies the formula,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{j^2 + 2j + 2} &= -\frac{1}{4} + \frac{1}{2(n^2 + 2n + 2)} \\ &+ \frac{2\pi}{e^{4\pi} - 1} \int_0^1 (e^{4\pi(1-u)} + e^{4\pi u}) \cos 2\pi(n+2)u \sin 2\pi n(1-u) \cot 2\pi(1-u) \, du \end{aligned}$$

So, it turns out the real problem is finding out the combination that yields the inverse polynomial that one is interested in.

4 Approaches based on cosine and sine

Though the exponential-based approach is probably more useful, if $\cos 2\pi(j+b) = \cos 2\pi b$ and $\sin 2\pi(j+b) = \sin 2\pi b$ are used as initial identities, different formulae can be obtained.

For cosine, starting from,

$$\cos 2\pi(j+b) = \cos 2\pi b = \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i)!},$$

which produces the recurrence equations,

$$\begin{cases} \frac{2(\sin \pi b)^2}{j+b} = 2\pi \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+2)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^2} = (2\pi)^2 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+2)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^3} = \frac{(2\pi)^2}{2!(j+b)} - (2\pi)^3 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+4)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^4} = \frac{(2\pi)^2}{2!(j+b)^2} - (2\pi)^4 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+4)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^5} = \frac{(2\pi)^2}{2!(j+b)^3} - \frac{(2\pi)^4}{4!(j+b)} + (2\pi)^5 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+6)!} \\ \frac{2(\sin \pi b)^2}{(j+b)^6} = \frac{(2\pi)^2}{2!(j+b)^4} - \frac{(2\pi)^4}{4!(j+b)^2} + (2\pi)^6 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+6)!} \\ \vdots \end{cases}$$

Likewise, for the sine,

$$\sin 2\pi(j+b) = \sin 2\pi b = \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+1)!},$$

the below recurrence equations are obtained,

$$\begin{cases} \frac{\sin 2\pi b}{j+b} = 2\pi \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+1)!} \\ \frac{\sin 2\pi b}{(j+b)^2} = \frac{2\pi}{j+b} - (2\pi)^2 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+3)!} \\ \frac{\sin 2\pi b}{(j+b)^3} = \frac{2\pi}{(j+b)^2} - (2\pi)^3 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+3)!} \\ \frac{\sin 2\pi b}{(j+b)^4} = \frac{2\pi}{(j+b)^3} - \frac{(2\pi)^3}{3!(j+b)} + (2\pi)^4 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+5)!} \\ \frac{\sin 2\pi b}{(j+b)^5} = \frac{2\pi}{(j+b)^4} - \frac{(2\pi)^3}{3!(j+b)^2} + (2\pi)^5 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i}}{(2i+5)!} \\ \frac{\sin 2\pi b}{(j+b)^6} = \frac{2\pi}{(j+b)^5} - \frac{(2\pi)^3}{3!(j+b)^3} + \frac{(2\pi)^5}{5!(j+b)} - (2\pi)^6 \sum_{i=0}^{\infty} \frac{(-1)^i (2\pi(j+b))^{2i+1}}{(2i+7)!} \\ \vdots \end{cases}$$

The formulae derived with these approaches break down for integer and half-integer b .

4.1 Cosine approach

Using the two systems of recurrence equations, we have more than one way to derive formulae for $HP_k(n)$. Unless we combine them, we have a very complicated recursion to determine the formula for $HP_k(n)$. Let us go over the simplest way.

Setting $a = 1$ without loss of generality, and intertwining the two systems of equations, the cosine-based $HP_k(n)$ formula is,

$$\sum_{j=1}^n \frac{1}{(j+b)^k} = -\frac{1}{2b^k} + \frac{1}{2(n+b)^k} - \frac{(2\pi)^k}{2} \int_0^1 p_k(u) (\cos 2\pi(n+b)u - \cos 2\pi bu) \cot \pi u \, du,$$

and below is the simplified recurrence equation on $p_k(u)$,

$$p_k(u) = \frac{(-1)^{\lfloor (k+1)/2 \rfloor}}{2 \sin \pi b} \left(\frac{(1-u)^{k-1}}{(k-1)!} + \sum_{j=1}^{\lfloor k/2 \rfloor} \frac{(-1)^j p_{2j-1}(u)}{(k-2j+1)!} \right) \cdot \begin{cases} -\frac{1}{\sin \pi b}, & \text{if } k \text{ is odd} \\ \frac{1}{\cos \pi b}, & \text{if } k \text{ is even} \end{cases}$$

To solve this equation, it is necessary to separate odd from even. For the odd one can make $q_k(u) = p_{2k+1}(u)$, and hence for $k \geq 0$,

$$q_k(u) = \frac{(-1)^k}{2(\sin \pi b)^2} \left(\frac{(1-u)^{2k}}{(2k)!} - \sum_{j=0}^{k-1} \frac{(-1)^j q_j(u)}{(2k-2j)!} \right)$$

The recurrence for even is very similar and is a function of $q_k(u)$. The generating functions of odd and even $p_k(u)$ are therefore,

$$f(x) = \frac{x \cos x(1-u)}{\cos x - \cos 2\pi b} \text{ and } g(x) = \sin x \frac{x \cos x(1-u)}{\cos x - \cos 2\pi b},$$

and the final formulae can be expressed as,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k+1}} &= -\frac{1}{2b^{2k+1}} + \frac{1}{2(n+b)^{2k+1}} \\ &\quad - \frac{(2\pi)^{2k+1}}{2} \int_0^1 \frac{f^{(2k+1)}(0)}{(2k+1)!} (\cos 2\pi(n+b)u - \cos 2\pi bu) \cot \pi u \, du, \text{ and} \end{aligned} \quad (14)$$

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k}} &= -\frac{1}{2b^{2k}} + \frac{1}{2(n+b)^{2k}} \\ &\quad - \frac{(2\pi)^{2k}}{2 \sin 2\pi b} \int_0^1 \left(\frac{(-1)^k (1-u)^{2k-1}}{(2k-1)!} + \frac{g^{(2k)}(0)}{(2k)!} \right) (\cos 2\pi(n+b)u - \cos 2\pi bu) \cot \pi u \, du \end{aligned} \quad (15)$$

4.2 Sine approach

Since the process for sine is analogous, only the final formulae are shown. Let $f(x)$ and $g(x)$ be the functions,

$$f(x) = \frac{x \sin x(1-u)}{\cos x - \cos 2\pi b} \text{ and } g(x) = \sin x \frac{x \sin x(1-u)}{\cos x - \cos 2\pi b}$$

The even and odd $HP_k(n)$ formulae can be expressed as a function of their derivatives by,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k}} &= -\frac{1}{2b^{2k}} + \frac{1}{2(n+b)^{2k}} \\ &+ \frac{(2\pi)^{2k}}{2} \int_0^1 \frac{f^{(2k)}(0)}{(2k)!} (\sin 2\pi(n+b)u - \sin 2\pi bu) \cot \pi u \, du, \end{aligned} \quad (16)$$

and,

$$\begin{aligned} \sum_{j=1}^n \frac{1}{(j+b)^{2k+1}} &= -\frac{1}{2b^{2k+1}} + \frac{1}{2(n+b)^{2k+1}} \\ &+ \frac{(2\pi)^{2k+1}}{2 \sin 2\pi b} \int_0^1 \left(\frac{(-1)^k (1-u)^{2k}}{(2k)!} + \frac{g^{(2k+1)}(0)}{(2k+1)!} \right) (\sin 2\pi(n+b)u - \sin 2\pi bu) \cot \pi u \, du \end{aligned} \quad (17)$$

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